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Hye Hyun Ma

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Current Academic Appointmets

Sejong University, Department of Media and Communication **Seoul, Korea**
Postdoctoral Researcher *Sep. 2026 - Present*

Hanyang University, Department of Advertising & PR **Ansan, Korea**
Lecturer *Sep. 2026 - Present*

Education

Hanyang University **Seoul, Korea**
Doctor of Philosophy in Advertising & Public Relations *Mar. 2021 - Aug. 2026*

- Dissertation: The Utility of Bayesian ANOVA as an Alternative Inferential Framework in the Social Sciences: A Monte Carlo Simulation Comparison with Frequentist ANOVA (Advisor: Prof. Byoungkwan Lee)

Hanyang University **Seoul, Korea**
Master of Arts in Advertising & Public Relations *Sep. 2018 - Feb. 2020*

- Thesis: Applying Bayesian Structural Time-series Models to Estimate the Effectiveness of Public Advertising: Causal Impact of Television Anti-Smoking Campaign on Tobacco Sales in Korea (Advisor: Prof. Byoungkwan Lee)

Hanyang University **Ansan, Korea**
Bachelor of Arts in Advertising & Public Relations *Mar. 2015 - Aug. 2018*

Research Interests

- **Computational Communication Science:** *Bayesian statistics and model-based inference for social science research; Causal inference and computational methods for complex communication data*
- **Meta-Science & Statistical Methodology:** *Statistical power, replicability, and evidence quality in communication research; Methodological decision-making and reporting practices in quantitative research*
- **Public Communication & Mediated Interaction:** *Public communication campaigns and their effects in health and risk contexts; Social interaction and user experiences in technology-mediated contexts (HCI/HAI/HRI)*

Refereed Journal Articles

✉ for a corresponding author.

1. Lee, B., Kim, E., **Ma, H. H.**, & Keum, S. (2026). Causal Inference of Public Communication Campaign Effects Using Machine Learning: Applying Causal Forests to Estimate Heterogeneous Treatment Effects. *Journal of Public Relations, 30*(3), 89-117. https://doi.org/10.15814/jpr.2026.30.3.89
2. Paek, H.-J., **Ma, H. H.**, & Hove, T. (2026). Preventing Excessive Lottery Behaviors by Correcting Cognitive Distortions: An Online Message Experiment Study. *The Social Science Journal.* Advance online publication. https://doi.org/10.1080/03623319.2026.2721309
3. Choi, M. Y., **Ma, H. H.**, & Oh, H. J. (2026). The Role of Risk Perception and Information-Seeking in Acceptance of Alternative Proteins. *Health Communication, 41*(9), 1507–1518. https://doi.org/10.1080/10410236.2025.2573778
4. **Ma, H. H.**, Lee, M., Kim, E., & Pang, S. (2026). The Effects of Virtual Social Touch on Interpersonal Communication Outcomes in Virtual Environments: A Meta-Analysis. *International Journal of Human–Computer Interaction, 42*(4), 2081–2097. https://doi.org/10.1080/10447318.2025.2526590
5. Paek, H.-J., & **Ma, H. H.**✉ (2025). Effects of Cognitive Distortions, Lottery-Related Social Relationships, and Media Campaign Exposure on Excessive Lottery Behavior: Implications for Responsible Lottery Campaign Strategies. *Journal of Public Relations*, *29*(4), 1-30. https://doi.org/10.15814/jpr.2025.29.4.1
6. Lee, B., **Ma, H. H.**✉, Kim, E., Pang, S., & Kim, T. (2025). Replicability of Communication Research in Korea: Evidence of Publication Bias and p-Hacking. *Korean Journal of Journalism & Communication Studies, 69*(4), 36-71. https://doi.org/10.20879/kjjcs.2025.69.4.002
7. **Ma, H. H.**, Choi, M. Y., & Paek, H.-J. (2025). Effectiveness of Foodborne Disease Prevention Campaign through Metaverse in South Korea: Integrating User Experience and Technology Acceptance Factors. *Health Communication Research, 24*(2), 121-144. https://doi.org/10.24172/hcr.2025.24.2.121
8. Hove, T., **Ma, H. H.**, & Paek, H. J. (2025). Using Webtoons for Public Service Communication: Factors Associated with Promoting Sound Lottery Behavior. *Journal of Digital Contents Society*, *26*(3), 635-644. http://dx.doi.org/10.9728/dcs.2025.26.3.635
9. Jeon, M., **Ma, H. H.**, Lee, M., & Shim, S. W. (2024). The Study on Content Analysis of Deceptive Advertising of Print Media since 2009. *The Korean Advertising & PR Practitioners Society*, *17*(4), 129-159. https://doi.org/10.21331/jprapr.2024.17.4.005
10. Lee, C. E., **Ma, H. H.**, & Paek, H.-J. (2024). How the Use of Environmental Information through Media Leads to Pro-environmental Behavior: Roles of Information Insufficiency and Information Gathering Capacity in Information Processing. *Korean Journal of Journalism & Communication Studies*, *68*(5), 196-232. https://doi.org/10.20879/kjjcs.2024.68.5.006
11. Oh, H. J., Lee, B., **Ma, H. H.**, Jang, D., & Park, S. (2024). A preliminary study for developing perceived ESG scale to measure public perception toward organizations' ESG performance. *Public Relations Review*, *50*(1), 102398. https://doi.org/10.1016/j.pubrev.2023.102398
12. **Ma, H. H.**, Keum, S., Pang, S., Oh, J., Lee, J., & Lee, B. (2023). The Influences of Nonverbal Communication Cues of Avatars on Self-Disclosure in Immersive Social VR: Focusing on Inter-Avatar Distance and Physical Attractiveness of the Interlocutor's Avatar. *Korean Journal of Journalism & Communication Studies, 67*(4), 47-88. https://doi.org/10.20879/kjjcs.2023.67.4.002
13. **Ma, H. H.**, & Lee, B. (2022). Applying Bayesian Structural Time-series Models to Estimate the Effectiveness of Public Advertising: Causal Impact of Television Anti-Smoking Campaign on

Tobacco Sales in Korea. *The Korean Journal of Advertising, 33*(8), 7-30. https://doi.org/10.14377/KJA.2022.11.30.7

14. **Ma, H. H.**, & Shim, S. (2019). The Study on Unjustice Indication and Advertising during 10 years since 2009. *Review of Culture and Economy, 22*(3), 117-152. https://doi.org/10.36234/kace.2019.22.3.117

Manuscripts Under Review & In Progress

1. **Ma, H. H.**, Hove, T., Lee, S., & Paek, H.-J. (Under review, R2). Promoting Responsible Gambling in South Korea: A Formative Social Marketing Study on Lottery Audience Segmentation
2. Kim, E., **Ma, H. H.**, Lee, B., Jeon, S., Levine, T. R., & Jang, H. (Under review, R3). Smartphone Overdependence and Suicidal Ideation Among Korean Adolescents in Care Services: Roles of Peer Relationships and Mental Health
3. Lee, B., **Ma, H. H.**, Paek, H.-J., Kim, T., & Hong, S. (Under review). Reliability Generalization Meta-Analysis of Health Belief Model Scales: Measurement Quality, Moderators, and Implications for Statistical Power
4. Pang, S., **Ma, H. H.**, Kim, E., & Lee, B. (Under review). Getting Closer and Feeling Worse? Effects of Inter-Avatar Distance, Avatar Attractiveness, and Gender on Interpersonal Outcomes in Immersive Virtual Reality
5. **Ma, H. H.** & Lee, B. (Under drafting). Statistical Power and Replicability in the Communication Literature in Korea: An Empirical Meta-Scientific Assessment of ANOVA Practices
6. **Ma, H. H.** & Lee, B. (Under drafting). Is Bayesian ANOVA a Useful Alternative for Social Scientists Under Assumption Violations? A Monte Carlo Simulation Study
7. **Ma, H. H.**, Paek, H.-J., & Lee, B. (In Preparation). Assessing P-Hacking and Replicability in Communication Research: Evidence from the Journal of Communication

Conference Presentations

1. **Ma, H. H.** & Lee, B. (2026, October). Statistical Power in the Communication Literature in Korea: An Empirical Meta-Scientific Assessment of ANOVA. Paper presented at Korean Society For Journalism And Communication Studies (KSJCS). Suwon, Korea.
2. Lee, B., Kim, E., **Ma, H. H.**, & Keum, S. Y. (2026, June). Causal Inference of Public Communication Campaign Effects Using Machine Learning: Applying Causal Forests to Estimate Heterogeneous Treatment Effects. Paper presented at Korea Health Communication Association (KHCA). Chuncheon, Korea.
3. **Ma, H. H.**, Lee, B., Paek, H.-J., Kim, T., & Hong, S. (2026, May). Reliability Generalization Meta-Analysis of Health Belief Model Scales: Measurement Quality, Moderators, and Implications for Statistical Power. Paper presented at Korean Society For Journalism And Communication Studies (KSJCS). Yeosu, Korea.
4. **Ma, H. H.** (2025, May). Exploring the Probabilistic Causal Structure of Media Trust: A Bayesian Network Analysis. Paper presented at Korean Association For Communication And Information Studies (KACIS). Busan, Korea.

5. **Ma, H. H.**, & Paek, H.-J. (2024, June). Designing Public Messages to Prevent Lottery Addiction*:* An Experiment Study to Correct Cognitive Errors. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.
6. **Ma, H. H.**, Kim, E., Pang, S. & Lee, B. (2024, May). Simulated touch: A Meta-Analysis of the Effects of Virtual Social Touch on Interpersonal Communication Outcomes. Paper presented at Korean Society For Journalism And Communication Studies (KSJCS). Gyeongju, Korea.
7. **Ma, H. H.**, Lee, S., & Paek, H.-J. (2024, April). Effective Public Communication on Responsible Lottery Purchase: Audience Segmentation Strategy. Paper presented at Korean Academic Society For Public Relations (KASPR). Jeonju, Korea.
8. Choi, M. Y., **Ma, H. H.**, & Oh, H. J. (2023, November). The Influences of Food Literacy on the Intention to use Alternative Proteins: Moderated Mediation Effects of Consumers' Information-seeking Behavior and Risk Perception. Paper presented at Korean Academic Society For Public Relations (KASPR). Seoul, Korea.
9. **Ma, H. H.**, Choi, M. Y., Oh, J., & Paek, H.-J. (2023, June). Exploring the Effectiveness of Health Campaigns Using Metaverse: A Case Study of the 'ZEPETO Dr. Jekyll's World' Food Poisoning Prevention Campaign. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.
10. Oh, H. J., Park, S., **Ma, H. H.**, Lee, B., & Jang, D. (2023, May). A Preliminary Study for Developing Perceived ESG Scale to Measure Public Perception Toward Organizations' ESG Performance. Paper presented at ICA-KACA research session. Toronto, Canada.
11. **Ma, H. H.**, Keum, S. Y., Pang, S., Oh, J. H., Lee, J., & Lee, B. (2023, May). The Impact of Nonverbal Communication Cues of Avatars on Self-Disclosure in an Immersive Social VR: Focusing on Inter-Avatar Distance and Physical Attractiveness of the Interlocutor's Avatar. Paper presented at Korean Society For Journalism And Communication Studies (KSJCS). Jeju, Korea.
12. Pang, S., **Ma, H. H.**, & Lee, B. (2023, May). Inter-Avatar Distance and Conversation Satisfaction in Immersive VR: Exploring the Moderating Effects of the Sense of Embodiment. Paper presented at Journal of Cyber communication Academic Society (JCAS). Seoul, Korea.
13. Lee, B., Park, S., Oh, H. J., Jang, D., **Ma, H. H.**, & Cho, J. H. (2022, November). Development of a Social Data-Based Perceived ESG Index for Public Perception Evaluation of Organizational ESG Activities. Paper presented at Korean Academic Society For Public Relations (KASPR). Seoul, Korea.
14. **Ma, H. H.**, & Lee, B. (2021, November). Causal Impact of Television Anti-Smoking Campaign on Tobacco Sales in Korea: An Application of Bayesian Structural Time-Series Models. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.
15. **Ma, H. H.**, & Lee, B. (2021, November). Estimating Causal Effects of Health Communication Campaigns. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.
16. **Ma, H. H.**, & Lee, B. (2021, June). Exploration of Evaluation Models for Government Advertising Effects. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.
17. **Ma, H. H.**, Shin, J., Joo, D., Mang, S., & Lee, B. (2018, December). A Study on the Framing Characteristics of Intimate Partner Violence Media Coverage. Paper presented at Korea Health Communication Association (KHCA). Seoul, Korea.

Research Experiences

PI

- **Evaluation of Human-Robot Interaction Based on User Experience** *Sep. 2024 - Aug. 2026*
  [Ph.D. Student Research Fellowship]
  Funder: National Research Foundation of Korea (NRF)
  Grant: KRW 40,000,000 (~USD 28,000)

Co-Investigator

- **Data-Driven Performance Analysis and Strategic Planning for Tobacco Control Policy Communications [PI: Prof. Hye-Jin Paek]** *Aug. 2026 - Nov. 2026*
  Funder: Korea Health Promotion Institute
  Grant: KRW 75,000,000 (~USD 50,000)

Research Assistant

- **Effectiveness Evaluation and Long-Term Communication Strategy for Lottery Promotion [PI: Prof. Hye-Jin Paek]** *Nov. 2023 - Mar. 2024*
  Funder: Donghaeng Lottery Co., Ltd
  Grant: KRW 120,000,000 (~USD 80,000)
- **Developing Plain Language Summary Guidelines for Scientific Information Communication in Outbreaks [PI: Prof. Myung-Il Choi]** *May. 2023 - Dec. 2023*
  Funder: Korea Disease Control and Prevention Agency
  Grant: KRW 90,000,000 (~USD 60,000)
- **Data Analysis for Validating P-ESG Measurement [PI: Prof. Byoungkwan Lee]** *Jun. 2022 - Jul. 2022*
  Funder: PR ONE Co., Ltd
  Grant: KRW 10,000,000 (~USD 7,000)

Teaching Experiences

Lecturer, Hanyang University

- APR 3044: Advanced Data Analytics *Fall 2026*
- APR 8025: Data Analytics (Graduate-level course) *Fall 2026*

Teaching Fellow (Instructor of Selected Sessions), Hanyang University

- APR 3026: Health Communication *Fall 2023, Fall 2024*
- APR 1027: Data Analytics *Spring 2024, Spring 2025, Spring 2026*
- APR 3044: Advanced Data Analytics *Fall 2025*

Teaching Assistant, Hanyang University

- APR 3026: Health Communication *Fall 2018, Fall 2019*
- APR 1027: Data Analytics *Spring 2018, Spring 2019*
- APR 613: Statistics in Advertising and PR (Graduate-level course) *Fall 2023, Fall 2024, Fall 2025*

Awards & Honors

- Outstanding Doctoral Dissertation Award, Hanyang University *2026*
  "The Utility of Bayesian ANOVA as an Alternative Inferential Framework in the Social Sciences: A Monte Carlo Simulation Comparison with Frequentist ANOVA"
- Best Paper Award, Korea Health Communication Association *2025*
  "Effectiveness of Foodborne Disease Prevention Campaign through Metaverse in South Korea: Integrating User Experience and Technology Acceptance Factors"
- Best Paper Award of the Year, Korean Society For Journalism And Communication Studies *2025*
  "Replicability of Communication Research in Korea: Evidence of Publication Bias and p-Hacking"
- Outstanding Graduate Student Paper Award, Korean Association For Communication And Information Studies (KACIS), Graduate Student Division *2025*
  "Exploring the Probabilistic Causal Structure of Media Trust: A Bayesian Network Analysis"
- Top Paper Award, International Communication Association (ICA), Korean American Communication Association (KACA) Research Session *2023*
  "A Preliminary Study for Developing Perceived ESG Scale to Measure Public Perception toward Organizations' ESG Performance"
- Humanities 100-Year Scholarship (formerly National Excellence Scholarship), Korea Student Aid Foundation (KOSAF), Ministry of Education *2017-2018*
  A prestigious national merit-based scholarship awarded to top-tier undergraduate students in humanities and social sciences, providing full tuition and a living stipend
- Bronze Prize, Undergraduate Research Program (URP), Hanyang University *2017*
  "Predicting the Future Performance of Hitters in the KBO (Korea Baseball Organization) League: A Proposal for a Measurement Model Using Data Science Methodologies"

Professional Services

Editorial & Academic Service

- Editorial Assistant, Health Communication Research *Feb. 2023 - Feb. 2024*

Conference & Journal Reviewer

- Korean Advertising and Public Relations Association (KADPR) Annual Conference *2026*

External Advisory

- Subject Matter Expert Consultation, Korea Health Promotion Institute (KHEPI) *2026*

Certifications & Certificated programs

- *Machine Learning for Causal Inference, The Korea Society of Management information Systems* *Aug. 2026*
- *Montpellier International Summer School, France* *Aug. 2017*

References

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(Additional references are available upon request)